

```
abdo@abdo: ~$ systemctl list-units --type=service --all
```

UNIT	LOAD	ACTIVE	SU
accounts-daemon.service	loaded	active	running
acpid.service	loaded	active	running
alsa-restore.service	loaded	active	exit
alsa-state.service	loaded	inactive	deleted
anacron.service	loaded	inactive	deleted
apparmor.service	loaded	active	exit
apport-autoreport.service	loaded	inactive	deleted
apport.service	loaded	active	exit
apt-daily-upgrade.service	loaded	inactive	deleted
apt-daily.service	loaded	inactive	deleted
● auditd.service	not-found	inactive	deleted
● auto-cpufreq.service	not-found	inactive	deleted
avahi-daemon.service	loaded	active	running
blk-availability.service	loaded	active	exit
bluetooth.service	loaded	active	running
colord.service	loaded	active	running
● connman.service	not-found	inactive	deleted
● console-screen.service	not-found	inactive	deleted
console-setup.service	loaded	active	exit
cron.service	loaded	active	running
cups-browsed.service	loaded	active	running
cups.service	loaded	active	running
dbus.service	loaded	active	running
dm-event.service	loaded	inactive	deleted
dmesg.service	loaded	inactive	deleted
dpkg-db-backup.service	loaded	inactive	deleted
e2scrub_all.service	loaded	inactive	deleted
e2scrub_reap.service	loaded	inactive	deleted
emergency.service	loaded	inactive	deleted
● fcoe.service	not-found	inactive	deleted
fstrim.service	loaded	inactive	deleted
fwupd-refresh.service	loaded	inactive	deleted
fwupd.service	loaded	active	running
gdm.service	loaded	active	running
getty-static.service	loaded	inactive	deleted
getty@tty1.service	loaded	inactive	deleted

lines 1-27 skipping

```
[1]~$ systemctl list-units --type=service --all
abdo@abdo:~$ systemctl get-default
graphical.target
abdo@abdo:~$ systemctl set-default multi-user.target
Created symlink /etc/systemd/system/default.target → /lib/systemd/system/multi-user.target.
abdo@abdo:~$ systemctl get-default
multi-user.target
abdo@abdo:~$ systemctl set-default graphical.target
```

: exit

```
abdo@abdo:~$ echo "Hello Root" | mail -s "Test Mail root" root
```

```
abdo@abdo:~$ sudo mail
```

"/var/mail/root": 6 messages 6 new

>N	1	Abdo Ibrahim	16/423	:14	28	ح	دیں	Test Mail
N	2	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	3	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	4	Abdo Ibrahim	15/384	:14	28	ح	دیں	Test Mail rott
N	5	root	22/374	:14	28	ح	دیں	Test Mail root
N	6	Abdo Ibrahim	14/375	:14	28	ح	دیں	Test Mail root

? ☐

```
abdo@abdo:~$ systemctl stop postfix
abdo@abdo:~$ echo "mail with postfix stopped" | mail -s "postfix test" root
abdo@abdo:~$ sudo mail
```

```
"/var/mail/root": 8 messages 8 new
```

>N	1	Abdo Ibrahim	16/423	:14	28	ح	دیں	Test Mail
N	2	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	3	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	4	Abdo Ibrahim	15/384	:14	28	ح	دیں	Test Mail rott
N	5	root	22/374	:14	28	ح	دیں	Test Mail root
N	6	Abdo Ibrahim	15/384	:14	28	ح	دیں	Test Mail root
N	7	Abdo Ibrahim	15/369	:14	28	ح	دیں	test root
N	8	Abdo Ibrahim	15/389	:14	28	ح	دیں	test root

```
? exit
```



```
abdo@abdo:~$ sudo systemctl start postfix
abdo@abdo:~$ echo "mail with postfix start" | mail -s "postfix test after restart" root
abdo@abdo:~$ mail root
```

Cc:
Subject:

Null message body; hope that's ok

```
abdo@abdo:~$ sudo mail
```

"/var/mail/root": 14 messages 14 new

>N	1	Abdo Ibrahiim	16/423	:14	28	ح	ديين	Test Mail
N	2	Abdo Ibrahiim	15/379	:14	28	ح	ديين	Test Mail
N	3	Abdo Ibrahiim	15/379	:14	28	ح	ديين	Test Mail
N	4	Abdo Ibrahiim	15/384	:14	28	ح	ديين	Test Mail rott
N	5	root	22/374	:14	28	ح	ديين	Test Mail root
N	6	Abdo Ibrahiim	15/384	:14	28	ح	ديين	Test Mail root
N	7	Abdo Ibrahiim	15/369	:14	28	ح	ديين	test root
N	8	Abdo Ibrahiim	15/389	:14	28	ح	ديين	test root
N	9	Abdo Ibrahiim	15/397	:14	28	ح	ديين	postfix test
N	10	Abdo Ibrahiim	15/398	:14	28	ح	ديين	postfix test
N	11	Abdo Ibrahiim	15/398	:14	28	ح	ديين	postfix test
N	12	Abdo Ibrahiim	15/398	:14	28	ح	ديين	postfix test
N	13	Abdo Ibrahiim	14/400	:14	28	ح	ديين	postfix test after restart



```
GNU nano 6.2 /tmp/crontab.sya14n/crontab *  
# Edit this file to introduce tasks to be run by cron.  
#  
# Each task to run has to be defined through a single line  
# indicating with different fields when the task will be run  
# and what command to run for the task  
#  
# To define the time you can provide concrete values for  
# minute (m), hour (h), day of month (dom), month (mon),  
# and day of week (dow) or use '*' in these fields (for 'any').  
#  
# Notice that tasks will be started based on the cron's system  
# daemon's notion of time and timezones.  
#  
# Output of the crontab jobs (including errors) is sent through  
# email to the user the crontab file belongs to (unless redirected).  
#  
# For example, you can run a backup of all your user accounts  
# at 5 a.m every week with:  
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/  
#  
# For more information see the manual pages of crontab(5) and cron(8)  
#  
# m h dom mon dow command  
  
*/10 8-17 * * * free -m | mail -s "Memory Status" root
```

abdo@abdo:~\$ sudo mail

"/var/mail/root": 14 messages 14 new

>N	1	Abdo Ibrahim	16/423	:14	28	ح	دیں	Test Mail
N	2	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	3	Abdo Ibrahim	15/379	:14	28	ح	دیں	Test Mail
N	4	Abdo Ibrahim	15/384	:14	28	ح	دیں	Test Mail rott
N	5	root	22/374	:14	28	ح	دیں	Test Mail root
N	6	Abdo Ibrahim	15/384	:14	28	ح	دیں	Test Mail root
N	7	Abdo Ibrahim	15/369	:14	28	ح	دیں	test root
N	8	Abdo Ibrahim	15/389	:14	28	ح	دیں	test root
N	9	Abdo Ibrahim	15/397	:14	28	ح	دیں	postfix test
N	10	Abdo Ibrahim	15/398	:14	28	ح	دیں	postfix test
N	11	Abdo Ibrahim	15/398	:14	28	ح	دیں	postfix test
N	12	Abdo Ibrahim	15/398	:14	28	ح	دیں	postfix test
N	13	Abdo Ibrahim	15/410	:14	28	ح	دیں	postfix test after rest
N	14	Abdo Ibrahim	14/351	:14	28	ح	دیں	

? exit


```
abdo@abdo:~$ sudo nano /etc/default/grub
```

```
abdo@abdo:~$ sudo update-grub
```



```
abdo@abdo:~$ sudo nano /etc/default/grub
abdo@abdo:~$ grep menuentry /boot/grub/grub.cfg
if [ x"${feature_menuentry_id}" = xy ]; then
    menuentry_id_option="--id"
    menuentry_id_option=""
export menuentry id option
menuentry 'Ubuntu' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-simple-e80c2e40-905d-4725-a805-10d4403dacdb' {
submenu 'Advanced options for Ubuntu' $menuentry_id_option 'gnulinux-advanced-e80c2e40-905d-4725-a805-10d4403dacdb' {
    menuentry 'Ubuntu, with Linux 6.8.0-90-generic' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-90-generic-advanced-e80c2e40-905d-4725-a805-10d4403dacdb' {
        menuentry 'Ubuntu, with Linux 6.8.0-90-generic (recovery mode)' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-90-generic-recovery-e80c2e40-905d-4725-a805-10d4403dacdb' {
            menuentry 'Ubuntu, with Linux 6.8.0-87-generic' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-87-generic-advanced-e80c2e40-905d-4725-a805-10d4403dacdb' {
                menuentry 'Ubuntu, with Linux 6.8.0-87-generic (recovery mode)' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-87-generic-recovery-e80c2e40-905d-4725-a805-10d4403dacdb' {
menuentry 'Windows Boot Manager (on /dev/sda4)' --class windows --class os $menuentry_id_option 'osprober-efi-F831-C8B0' {
menuentry 'UEFI Firmware Settings' $menuentry_id_option 'uefi-firmware' {
abdo@abdo:~$ sudo nano /etc/default/grub
abdo@abdo:~$
```

GNU nano 6.2

/etc/default/grub *

```
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'
```

```
GRUB_DEFAULT=0
```

```
GRUB_TIMEOUT_STYLE=menu
```

```
GRUB_TIMEOUT=20
```

```
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
```

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
```

```
GRUB_CMDLINE_LINUX=""
```



abdo@abdo: ~



```
GNU nano 6.2 /tmp/crontab.DGcF3Z/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
MAILTO=abdo
*/10 8-17 * * * free -m
```


abdo@abdo:~\$ sudo mail

"/var/mail/root": 14 messages 14 new

>N	1	Abdo Ibrahim	16/423	:14	28	ح	دعیں	Test Mail	
N	2	Abdo Ibrahim	15/379	:14	28	ح	دعیں	Test Mail	
N	3	Abdo Ibrahim	15/379	:14	28	ح	دعیں	Test Mail	
N	4	Abdo Ibrahim	15/384	:14	28	ح	دعیں	الحقیقۃ المیل مجاش معرفش المشکله عشان	
N	5	root	22/374	:14	28	ح	دعیں	انا علی اوبنتو ولا ایه	
N	6	Abdo Ibrahim	15/384	:14	28	ح	دعیں	Test Mail root	
N	7	Abdo Ibrahim	15/369	:14	28	ح	دعیں	test root	
N	8	Abdo Ibrahim	15/389	:14	28	ح	دعیں	test root	
N	9	Abdo Ibrahim	15/397	:14	28	ح	دعیں	postfix test	
N	10	Abdo Ibrahim	15/398	:14	28	ح	دعیں	postfix test	
N	11	Abdo Ibrahim	15/398	:14	28	ح	دعیں	postfix test	
N	12	Abdo Ibrahim	15/398	:14	28	ح	دعیں	postfix test	
N	13	Abdo Ibrahim	15/410	:14	28	ح	دعیں	postfix test after rest	
N	14	Abdo Ibrahim	14/351	:14	28	ح	دعیں		

? exit



abdo@abdo: ~



```
#!/bin/bash
```

```
if [ $# -eq 0 ]; then
    echo "script executed without arguments" >> /var/log/scriptlogs
else
    echo "Number of arguments: $#"
```

```
fi
```

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```

GNU nano 6.2 /etc/rsyslog.conf *

```
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf
```

```
#####
#### MODULES ####
#####
```

```
module(load="imuxsock") # provides support for local system logging
#module(load="immark") # provides --MARK-- message capability
```

```
# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")
```

```
# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")
```

```
# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")
```

```
*.* @logging_server_ip:514
```

```
#####
#### GLOBAL DIRECTIVES ####
#####
```

```
#
# Use traditional timestamp format.
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify


```
abdo@abdo:~$ sudo nano /etc/rsyslog.conf
abdo@abdo:~$ systemctl restart rsyslog
abdo@abdo:~$ sudo nano /etc/rsyslog.conf
abdo@abdo:~$ sudo nano /etc/rsyslog.conf
abdo@abdo:~$ systemctl restart rsyslog
abdo@abdo:~$ logger "Hello from workstation"
abdo@abdo:~$ sudo tail -f /var/log/syslog
```

بردو المفروض تظهر هنا

```
Dec 28 15:10:28 abdo rsyslogd: cannot resolve hostname 'logging_server_ip': Resource temporarily unavailable [v8.2112.0 try https://www.rsyslog.com/e/2027 ]
Dec 28 15:10:28 abdo rsyslogd: action 'action-0-builtin:omfwd' suspended (module 'builtin:omfwd'), next retry is Sun Dec 28 15:10:58 2025, retry nbr 0. There
should be messages before this one giving the reason for suspension. [v8.2112.0 try https://www.rsyslog.com/e/2007 ]
Dec 28 15:10:37 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151037.343652:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=48, type=200
Dec 28 15:10:39 abdo abdo: Hello from workstation
Dec 28 15:10:45 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151045.670610:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=48, type=205
Dec 28 15:10:53 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151053.992020:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=48, type=200
Dec 28 15:11:02 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151102.285629:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=44, type=205
Dec 28 15:11:02 abdo rsyslogd: cannot resolve hostname 'logging_server_ip': Resource temporarily unavailable [v8.2112.0 try https://www.rsyslog.com/e/2027 ]
Dec 28 15:11:10 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151110.107423:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=80, type=201
Dec 28 15:11:18 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151118.439277:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=48, type=205
Dec 28 15:11:59 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151159.400363:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=44, type=205
Dec 28 15:12:02 abdo gnome-shell[1905]: Can't update stage views actor <unnamed>[<MetaWindowGroup>:0x5a20aa2ae330] is on because it needs an allocation.
Dec 28 15:12:02 abdo gnome-shell[1905]: Can't update stage views actor <unnamed>[<MetaWindowActorX11>:0x5a20abbd9ad0] is on because it needs an allocation.
Dec 28 15:12:02 abdo gnome-shell[1905]: Can't update stage views actor <unnamed>[<MetaSurfaceActorX11>:0x5a20ac0e54e0] is on because it needs an allocation.
Dec 28 15:12:07 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151207.637835:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unpr
otect RTCP packet: size=80, type=201
```



```
GNU nano 6.2 /etc/rsyslog.conf *
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf

#####
#### MODULES ####
#####

#module(load="imuxsock") # provides support for local system logging
#module(load="immark") # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
#module(load="imklog" permitnonkernelfacility="on")

*. * @logging_server_ip:514

#####
#### GLOBAL DIRECTIVES ####
#####

#
# Use traditional timestamp format.
# To enable high precision timestamps, comment out the following line.
#
$ActionFileDefaultTemplate RSYSLOG_TraditionalFileFormat

# Filter duplicated messages

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```



```
abdo@abdo:~$ sudo systemctl restart rsyslog
abdo@abdo:~$ logger "Test message only to central server"
abdo@abdo:~$ sudo tail -f /var/log/syslog
Dec 28 15:15:34 abdo gnome-shell[1905]: Window manager warning: Overwriting existing binding of keysym 36 with keysym 36 (keycode f).
Dec 28 15:15:34 abdo gnome-shell[1905]: Window manager warning: Overwriting existing binding of keysym 37 with keysym 37 (keycode 10).
Dec 28 15:15:34 abdo gnome-shell[1905]: Window manager warning: Overwriting existing binding of keysym 39 with keysym 39 (keycode 12).
Dec 28 15:15:48 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151548.677918:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unprotect RTP packet: size=44, type=205
Dec 28 15:15:56 abdo dbus-daemon[1154]: [session uid=1000 pid=1154] Activating via systemd: service name='org.freedesktop.Tracker3.Miner.Extract' unit='tracker-extract-3.service' requested by ':1.111' (uid=1000 pid=2386 comm="/usr/libexec/tracker-miner-fs-3 " label="unconfined")
Dec 28 15:15:56 abdo systemd[1085]: Starting Tracker metadata extractor...
Dec 28 15:15:56 abdo dbus-daemon[1154]: [session uid=1000 pid=1154] Successfully activated service 'org.freedesktop.Tracker3.Miner.Extract'
Dec 28 15:15:56 abdo systemd[1085]: Started Tracker metadata extractor.
Dec 28 15:15:56 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151556.971328:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unprotect RTP packet: size=48, type=205
Dec 28 15:16:05 abdo teams-for-linux_teams-for-linux.desktop[3600]: [3600:1228/151605.265430:ERROR:third_party/webrtc/pc/srtp_transport.cc:157] Failed to unprotect RTP packet: size=48, type=205
```